Experiment 3: Factorial (Recursive vs Iterative)

Aim

To implement the **Factorial** of a number using both **Iterative** and **Recursive** methods in Python and compare the results.

Algorithm

Iterative Method

Step 1: Start.

Step 2: Read the number n.

Step 3: Initialize fact = 1.

Step 4: Multiply fact by each number from 1 to n.

Step 5: Display the factorial.

Step 6: Stop.

Recursive Method

Step 1: Start.

Step 2: Read the number n.

Step 3: If n is 0 or 1, return 1.

Step 4: Otherwise return n × factorial(n-1).

Step 5: Display the factorial.

Step 6: Stop.

Source Code (Python)

main.py Share Run

```python
# Factorial using Iterative and Recursive Methods

def recursive_fact(n):
    if n == 0 or n == 1:
        return 1
    return n * recursive_fact(n - 1)

n = int(input("Enter a number: "))

# Iterative
fact = 1
for i in range(1, n + 1):
    fact *= i

print("Iterative Factorial =", fact)
print("Recursive Factorial =", recursive_fact(n))
```

Output

Output

```
Enter a number: 5
Iterative Factorial = 120
Recursive Factorial = 120

=== Code Execution Successful ===
```

Time Complexity

Method	Time Complexity	Space Complexity
Iterative	O(n)	O(1)
Recursive	O(n)	O(n)

Result

The factorial of a given number was successfully computed using both Iterative and Recursive methods. Both methods produced the same result, verifying the correctness of the implementation.